

# Handover: the 2026-08-25 → 2026-08-26 arc

Written 2026-08-26 for a successor session. `handover.md` remains the standing session entry point and `HANDOFF.md` the deep research context; this file covers only what happened in this arc, why, and where it landed. Read it after those two.

## 0 What this arc was

It began as a request to explain the practical-equivalence ROPE used to score the crash-causation study, and turned into three things: a substantial revision of how that study's statistics are computed and reported, a diagnosis of why its metrics disagreed with each other, and the opening of a new workstream on human data that arrived mid-arc.

Nothing about the study's **findings** was overturned. What changed is the machinery used to express them, and in two places the machinery had been wrong.

## 1 Decisions taken, with their reasoning

These are settled unless someone reopens them. Each has a documented argument; the third column says where.

| # | decision                                                                                                                                                                                         | where argued                                                  |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| 1 | <b>The QUADRI5 5 000 are the target population, not a sample.</b> Only the synthetic side carries sampling error. Claims must name the ensemble and must not generalize to the fleet.            | docs/equivalence_rope_note.md §2.5                            |
| 2 | <b>ROPE thresholds are derived, not inherited:</b> $\theta_{thd} = 0.188$ , $\Theta_{thd} = 0.089$ , from a stated 10% tolerance on the injury-weighted mean.                                    | §3.1, §5, §5.2                                                |
| 3 | <b>Uniform bin weights (<math>\omega = 1</math>), no system under assessment.</b> Building an AEB to generate weights would not help; a very effective system produces degenerate weights.       | §4.3                                                          |
| 4 | <b>Severity stays operationalized as injury risk</b> , not relative speed at impact, with relative speed reported alongside as the assumption-free check.                                        | §5.3                                                          |
| 5 | <b>N = 5 bins</b> , against Eq. 4's prescribed 20, because 20 is unusable at this reference size and $\theta$ 's upward bias grows with N.                                                       | §2.4                                                          |
| 6 | <b>Comparisons between conditions are made as paired differences</b> , not by comparing marginal intervals.                                                                                      | docs/crash_causation_results.md §4.3b-ii                      |
| 7 | <b>The weighted aggregate is reported alongside <math>\theta</math> and <math>\Theta</math></b> , because neither bounds it.                                                                     | §3.2; implemented in <code>equivalence.aggregate_table</code> |
| 8 | <b>Fixed-clip (Randomized) data is the primary CZB fitting target</b> ; button-press is validation only, because of satisficing; Sequence is excluded for autocorrelation.                       | docs/czb_study1_data_plan.md §2(b)                            |
| 9 | <b>Comfort and dread boundaries fitted as an ordered model</b> , with the dread level well identified by "hard braking expected" and the comfort level treated as latent with a wider posterior. | docs/handbook/11_path_to_czb.md                               |

## 2 Two defects found in our own code

Both were ours, not the method's. Both are fixed; both required regenerating outputs.

**2.1 The weighted bootstrap understated uncertainty by about a factor of two.** `equivalence_test` drew reference *values* with probability proportional to weight and then treated the resample as unweighted, which claims the precision of 5 000 independent draws. The QUADRI5 crash weights are concentrated enough (top 10% of scenarios carry 54% of the weight) that the reference's effective sample size is **950**. Fixed by adding a resample parameter with three schemes — "population" (default), "cases", "values" (legacy) — in `src/equivalence/test.py`. Point estimates were unaffected; every HDI published before 2026-08-26 was too narrow.

**2.2 The assessment used N = 5 bins where the method's own rule gives 20.** `run_quadris.assess` hardcoded `n_bins=5`, correct when the reference was 100–200 seeds and never revisited at full population. Quantified rather than simply fixed, because the sweep showed N = 20 is unusable here: `replication/causation/bin_sensitivity.py`, results in `docs/crash_causation_results.md` §4.4.

## 3 Findings

**3.1 The ROPE was unpassable, for reasons unrelated to the model.** A null calibration — comparing the reference with itself — showed the original configuration could not be passed by a model that is exactly right. Three causes compounded: uniform weights applied a tolerance anchored to a 2% injury-risk baseline that our seeds never approach; the bootstrap defect above; and treating the reference as a sample. Settling decision 1 drops the noise floor from 0.177 to about **0.021**, at which point the verdicts describe the model.

**3.2 Choosing the bin-weight anchor P0 is choosing  $\theta$ thd.** They are one degree of freedom, not two, since P0 rescales every weight by a constant and  $\theta$  is linear in  $\omega$ . Only the weight *shape* is independent, and on our data it moves  $\theta$  by 7%. This is why decision 3 is coherent with decision 2 rather than an independent choice. (equivalence\_rope\_note.md §4.1)

**3.3  $\theta$  and  $\Theta$  do not bound the weighted aggregate.** They constrain only how mass is allocated between bins; the outer bins are open-ended. Condition B's actual mean injury risk is +12.6% while its binned approximation is -2.7%, the gap sitting almost entirely in the top band, where B's crashes average 0.0213 against the reference's 0.01615. (§3.2)

**3.4 Severity and response timing are near-independent in this population.** Conditions B and C differ 4.7-fold in crash rate yet their severity quartiles agree to 0.3 m/s; 71% of impact-speed variance is between-seed. Response timing decides *how many* crashes happen; the scenario and whether the driver brakes at all decide *how hard*. Consequence worth carrying: **a driver model validated only on crash severity distributions is close to unconstrained in its response timing.** (docs/severity\_vs\_timing.md)

**3.5  $\theta$  misreads three of the four metrics.** It assumes the reference can be cut into N equal pieces. Severity is smooth, so  $\theta = 0.148$  is meaningful.  $t_{nr}$  lives on a 0.05 s lattice with 25 distinct reference values.  $a_{f,min}$  carries a 48% atom at zero that collapses two of five quantile edges onto the same point, producing  $\theta = 1.058$  mechanically. On interpretable bins the non-braking share agrees to 4.6%, and on the weighted mean condition B is within **0.3%** while the CBM control over-brakes by **35%** — a far sharper separation than  $\theta$  ever showed.

**3.6 The B-versus-C ordering is statistically solid.** With the corrected wider intervals the marginal HDIs overlap, which would naively suggest no separation. As a paired difference with a shared reference draw:  $\theta_C - \theta_B = 0.061$ , 95% HDI [0.037, 0.107], same sign on every resample. (crash\_causation\_results.md §4.3b-ii) *[Corrected 2026-08-27: that analysis held both synthetic sides fixed and overstated the certainty. With the synthetic sides resampled,  $P(\theta_C > \theta_B) \approx 0.97$  and the HDI grazes zero — strong support, not unanimity. Committed script: replication/causation/paired\_difference.py; revised text in results doc §4.3b-ii.]*

**3.7 Where the study now stands.** Neither condition reaches practical equivalence at a defensible tolerance — B misses at 10% on its interval's upper end, C on every reading. The gain from this arc is that the miss is now a statement about the model rather than about the instrument.

## 4 The new data, and the cut-in workstream

**What arrived.** external/01\_studies/ — a crowdsourced clip-rating and button-press study, 80 participants, four scenarios, 67 stimulus kinematic traces at 10 Hz. Described in the new handbook appendix, chapter 15. Jonas's own CZB position paper arrived alongside at external/QUADRARUM/Bärgman\_On\_CZB\_as\_basis\_for\_ADAS\_adaptation 20260501.docx.

**Three data-quality findings not in its documentation**, all verified against the traces: Acceleration\_mps2 is unsigned and its definition could not be established (do not use it; differentiate Speed\_mps); the first and last frames are simulator teardown artifacts (~8.5% of rows); truck bounding boxes are broken (Length\_m = 0).

**Why the dataset suits the method.** The ego vehicle never responds — it holds a rigorously constant speed through the cut-in clips — so the comfort-zone field along a clip is a fixed function of time, computable once per stimulus and reused for every participant. The fixed-clip design samples the response-time **CDF** at six known truncation points; the button design samples the **density**. One accumulator predicts both.

**A cross-paradigm check worth knowing about.**  $P(\text{intervene} \mid \text{clip ends at } T)$  should equal  $P(\text{press} \leq T)$ . It holds within 0.06 at high criticality and shows a systematic excess of early pressing elsewhere, up to +0.26 at intermediate criticality — matching Jonas's documented concern about satisficing in the button paradigm, and now quantified.

**The cut-in scenario is started, not finished.** `src/comfortzone/cutin.py` loads and conditions the traces, assigns vehicle roles geometrically, and computes candidate predictors. Running it produced the arc's most useful negative result: **with released parameters the model does not discriminate TTC4 from TTC8 at all**. Decomposing the six preference terms shows why — the speed term sits at a constant  $-483$  because the default desired speed is 15 m/s against a 30.5 m/s stimulus (a staging error, fix by setting it from the clip), and the safety term is a saturated binary step, identical at a 10 m and a 21 m gap, because both cut-ins end under 0.75 s headway. Only the collision term discriminates, and it is two orders of magnitude smaller.

**The open design question**, and the thing to settle before any fitting: lane entry must become continuous. Proposed in handbook ch. 04 — lateral **overlap fraction** for the present state, and for the predictive half the comparison of **time-to-lateral-overlap** against the longitudinal TTC, since a collision requires lateral overlap to have begun by the time the gap closes. This also explains why lane-change *duration* was not a useful predictor in the QUADRARUM group's own analyses: duration is a property of the manoeuvre alone, whereas what matters is the lateral closing rate relative to the longitudinal one.

## 5 The CZB framing that was agreed

Recorded because Jonas confirmed it is the intended direction, and because it is falsifiable.

The conventional route estimates a percentile of an *observable* and inherits a multivariate problem — each scenario has its own observables, and combining marginal percentiles under a conjunction rule silently tightens the criterion (Jonas's own document works this through and proposes an elliptical joint-percentile fix). The active-inference route estimates a per-driver **scalar** level on the preference field, which has already absorbed speed, gap, closing rate and lateral position into one number, so the population distribution is one-dimensional by construction.

**The cross-scenario transfer test is what falsifies this:** fit the level on one scenario, predict the other three unchanged. **If transfer fails, the fallback is the elliptical formulation** — tracked deliberately.

A conceptual convergence worth keeping: Jonas's document defines the CZB as the point where "the satisficing condition is violated ... and behavioral regulation becomes necessary", which is very close to what the preference function *is* — a specification of acceptable outcomes rather than an optimum.

The fitted level absorbs response bias, paradigm offset and motor latency as well as the boundary. The design contains contrasts for most of these (the pre-onset condition identifies response bias; the counterbalanced framing contrast identifies the framing shift; the clip-versus-button contrast identifies the satisficing offset), but **motor latency is not identified at all** and the paradigm offset can never be a clean modality effect because Random always preceded Button. Documented with the open design decisions in `docs/handbook/11_path_to_czb.md`.

## 6 What changed on disk

**New documents** (each with Word and PDF):

- `docs/equivalence_rope_note.md` — the ROPE calibration and the project's settled configuration. Written to be sendable to Jian Wu; §5.1 is a paper-ready methods paragraph and §6 lists questions for the authors.
- `docs/severity_vs_timing.md` — why severity matches while timing and braking do not.
- `docs/czb_study1_data_plan.md` — the new dataset and the proposed CZB programme.
- `handover_2026-08-26.md` — this file.

**New handbook material** (revision round `{{R3}}`, dark green):

- ch. 04 — "The three files, line by line", "What a cut-in scenario would take", "The cut-in's open problem", plus the flow chart `figures/scenario_diff.png`
- ch. 08 — "How strict is practically equivalent? Reading a ROPE"
- ch. 11 — "From one driver's boundary to a population percentile", "What the fitted boundary level absorbs", and Levels 1–2 on the hierarchical model
- ch. 15 — **new appendix: the datasets**, one subsection per dataset with its caveats

**New code:**

| file                                  | what                                                                |
|---------------------------------------|---------------------------------------------------------------------|
| <code>src/comfortzone/cutin.py</code> | cut-in trace loading, manoeuvre-progress norm, candidate predictors |

| file                                              | what                                                                           |
|---------------------------------------------------|--------------------------------------------------------------------------------|
| src/equivalence/test.py                           | resample parameter (three schemes); aggregate diagnostics on EquivalenceResult |
| src/equivalence/report.py                         | aggregate_table                                                                |
| replication/causation/bin_sensitivity.py          | bin-count sweep from stored outputs                                            |
| replication/causation/severity_vs_timing.py       | per-bin diagnostics, atom structure, timing sensitivity                        |
| replication/causation/make_dissociation_figure.py | fig_dissociation.png, fig_metric_shapes.png                                    |
| docs/handbook/make_scenario_diagram.py            | figures/scenario_diff.png                                                      |

**Regenerated:** summary\_fullp\_abn.md, summary\_bin\_sensitivity\_fullp\_abn.md, and the affected sections of crash\_causation\_results.md. The 100-seed sensitivity ladder and the three non-abnormal rows of §4.3 still carry pre-correction intervals and are marked as such.

## 7 Where I would pick up

- 1 **Settle the continuous lane-entry form** (§4 above). Everything in the cut-in workstream waits on it, and it is a modification to the authors' preference function rather than a parameter choice, so it needs an explicit argument.
- 2 **Fix the desired-speed staging** in the cut-in predictors — set it from the clip rather than the default 15 m/s.
- 3 **Then fit:** boundary level on the Randomized clip-based binary surface, ordered model for comfort and dread, per-driver random effects; validate against button press times.
- 4 **Then transfer:** fit on cut-in, predict LTAP, cyclist overtake and truck overtake.
- 5 Standing items unchanged: Jonas's handbook review, the Jian Wu email (docs/email\_jian\_draft.md, now with the ROPE note as a companion), the a\_f,min execution model, and the unexercised epistemic half of the gaze system.

## 8 Git state

This arc is committed as a **single commit** on main, immediately after b689da1 ("Rewrite handover.md for the 2026-08-23 to -25 arc..."), which is where the previous arc finished. One commit rather than a series, because the pieces are interdependent: the bootstrap fix, the threshold change and the regenerated outputs are only coherent together, and splitting them would leave intermediate commits whose numbers contradict their own documentation. The commit covers modified README.md, handover.md, docs/crash\_causation\_results.md, handbook chapters 04/08/11, build\_handbook.py, run\_quadris.py and the three src/equivalence/ modules, plus the new documents, figures, scripts and handbook chapter 15 listed in section 6.

**A caveat about diffing against it.** Because the arc is one commit, a successor comparing "before" and "after" will see the *net* effect and not the sequence of corrections. Two of those corrections matter and are easy to miss in a net diff — the weighted-bootstrap defect (§2.1) and the threshold change from Wu's inherited 0.10/0.05 to the derived 0.188/0.089. Both are documented in docs/equivalence\_rope\_note.md; neither is visible as a separate step in the history.

One housekeeping note. The large cond\_\*\_fullp\*.csv files remain gitignored and must be regenerated from the commands logged in replication/causation/out/ before any assessment can be re-run. Everything else in docs/ is current: all markdown, Word and PDF builds were regenerated at the end of the arc, and no versioned -vN intermediates survive.

## 9 Caveats on this file

Written by the session that did the work, so it inherits that session's blind spots. Where it disagrees with the repository, trust the repository. Two specific things a successor should verify rather than assume: that the regenerated intervals in crash\_causation\_results.md §4.3 match replication/causation/out/summary\_fullp\_abn.md (they were copied by hand), and that the cut-in module's role assignment behaves on the truck and LTAP traces, which it has not been tested against.